

TUTTE ALGEBRAS OF GRAPHS AND FORMAL GROUP THEORY

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1. *Introduction*

In 1947, in a beautiful and influential paper [15], Tutte introduced a powerful algebraic framework for constructing invariants of multigraphs. This has spread increasing ripples down the years, and is, for example, a forerunner to certain recent exciting developments in knot theory [5].

However, combinatorialists have tended to play down the algebraic components of Tutte's work, and concentrate instead on the graph theory. Indeed, in the process of improving and extending his original ideas, Tutte himself [16] has felt the need to present the material 'cleared of algebraic superstructure'. It is our contention that this superstructure is actually remarkably interesting, and that part of it exposes links between combinatorics and the theory of formal groups, and thereby also with the algebraic topology of spaces such as loops on the 3-sphere and infinite-dimensional complex projective space [10].

Our aim here is to concentrate on one aspect of Tutte's ring, which is polynomial in infinitely many variables over $\mathbb{Z}$, by restricting attention to *simple* graphs. This reduces the number of variables to one, but has the attraction that we may enrich the structure in a natural and appealing way to that of a Hopf algebra supporting a difference operator (actually the discrete derivative); in this procedure, a graph becomes identified with its chromatic polynomial. These topics are the subject matter of our § 1 and § 2.

We then generalise so as to include recent developments in umbral calculus [8], which acts as the bridge with the theory of formal groups. This involves incorporating into the algebraic construction information concerning partitions of the vertices of the graphs, as explained in § 3 and § 4. There again results a polynomial algebra in one variable, but this time over a larger ring Φ_* . Once more we can naturally impose the structure of a Hopf algebra, which now supports an *umbral* difference operator, and a graph becomes identified with its *umbral* chromatic polynomial, as introduced in [12].

We are thus able in § 5 to establish a combinatorial interpretation for an arbitrary formal group over any ring free of additive torsion. This of course includes the universal case, and raises several interesting questions. We save a discussion of the interactions with algebraic topology for the future.

We refer the reader to Aigner [1] for general combinatorial terminology, to Sweedler [14] for all information concerning Hopf algebras, and to Hazewinkel [4] for an encyclopaedic description of the theory of formal groups.

It is a pleasure to acknowledge enjoyable and fruitful discussions with several mathematicians during the development of this material. In particular, Bill Schmitt provided privileged access to his ongoing work on combinatorial Hopf algebras, and so helped inspire § 2; we are currently working together on a sequel

more directly related to the combinatorial interpretation of formal group theory. Dave Benson and Dominic Welsh offered suggestions and encouragement during an Oxford seminar, whilst Colin Wright provided pertinent criticism throughout his period of support by SERC. And finally, David Carlisle and Bill Mitchell gave indispensable assistance in formulating the ideas in a context susceptible to generalisation; any future work extending to Tutte's infinity of variables will be joint with them.

2. The Hopf algebra of simple graphs

In our first two sections we describe a version of Tutte's original construction as restricted to simple graphs; by definition these have no loops, and at most one edge between any pair of vertices. It is helpful to begin by introducing some notation and preliminary results.

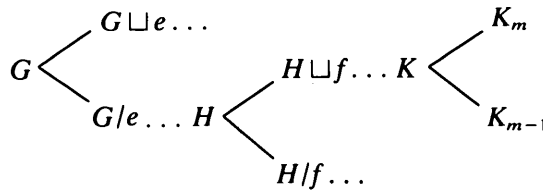
Given a simple graph G , we denote the sets of vertices and edges by $V = V(G)$ and $E = E(G)$ respectively, and the set of *non-edges*, or unordered pairs of vertices not in E , by $N = N(G)$. Unless otherwise stated, we always assume that $|V| = n$. We write $\Pi(V)$ for the poset of partitions of the vertices, ordered by refinement. We reserve the notation K_n for the *complete* graph on n vertices, and N_n for its complement the *null* graph, which has no edges. Thus $K_0 = N_0$ is the *empty* graph $\emptyset$, which has neither edges nor vertices.

For a subset $W \subseteq V$, and when V is clearly understood, we abbreviate the complement $V \setminus W$ to W' . We then write $G|_W$ for the induced subgraph, and given an element π of the poset $\Pi(V)$ we write $G|_\pi$ for the union of the subgraphs induced by each of the blocks. A *colour partition* of G is a partition of $V(G)$ into blocks which contain no edges.

For any non-edge e we write $G \sqcup e$ and G/e for the graphs obtained from G by respectively adjoining and then contracting e . Since we wish to remain within the category of simple graphs, we decree that contraction creates no multiple edges.

These ideas lead naturally to the addition-contraction tree $\mathcal{T}(G)$ of G . This has G itself as initial node, and each node H has successors $H \sqcup f$ and H/f for some arbitrary choice of non-edge f in $N(H)$. Thus $\mathcal{T}(G)$ is finite and its leaves are complete graphs $K_m = K_m(G)$. It may be represented as follows:

DIAGRAM 2.1.



Note that each m lies between 1 and n . For every node H we may construct a partition $\pi(H)$ of $V(G)$ by labelling each vertex v of H with the set of those vertices of G which are identified in creating v , and then choosing the labels as blocks.

The arbitrary choice of non-edge at each stage ensures that $\mathcal{T}(G)$ is not usually unique. In spite of this, it can be used to read off important information concerning G , as we now show by recalling a result of [11].

PROPOSITION 2.2. *As K_m ranges over the leaves of $\mathcal{T}(G)$, the partitions $\pi(K_m)$ range over the set of colour partitions of G .*

Proof. Suppose we are given a colour partition π of G . Starting at the initial node G , consider the non-edge $e = uv$ over which addition–contraction takes place. If u and v are in the same block of π , we modify π to be a colour partition of G/e , whilst if u and v are in different blocks of π , we retain π as a colour partition of $G \sqcup e$. Continuing in this fashion, we are led uniquely to a leaf K_m for which $\pi(K_m)$ is π .

Clearly, two different colour partitions will terminate at different leaves, so it only remains to prove that every leaf is reached by the above process. This follows from the fact that every $\pi(K_m)$ is a colour partition of G by construction.

By virtue of Proposition 2.2, the leaves of $\mathcal{T}(G)$ constitute a well-defined set of complete graphs, which it is often convenient to write as $\mathcal{L}(G)$.

In what follows, we shall only assume a graph to be defined up to isomorphism; in other words G stands for its equivalence class. This abuse of notation is commonplace, and often appears without comment in the literature. Its effect is to ensure that the set of *all* graphs on at most n vertices is finite, for any integer n .

Consider the graded polynomial algebra $k_* = \mathbb{Z}[u]$ on a single generator u . For reasons related to our later applications we assume that u has *complex* dimension 1, and therefore is of grading 2. Now take the free module $k_*\{G\}$ generated by all graphs G (including $\emptyset$), and let it be graded by supposing that G has complex dimension equal to the cardinality $|V(G)|$ of its vertex set. We propose to impose certain relations on $k_*\{G\}$ involving addition and contraction of non-edges.

Thus we let R_* denote the submodule generated by all terms of the form

$$G - G \sqcup e - uG/e \quad \text{with } e \in N(G), G \neq \emptyset, \quad (2.3)$$

and write T_* for the quotient module. The notation is designed to suggest that T_* is also graded, as follows from observing that the terms in (2.3) are all homogeneous. On occasion, we shall denote the equivalence class of a graph G in T_* simply by G .

To determine the structure of T_* it is helpful to introduce an alternative module of relations Q_* , generated by all terms of the form

$$G - \sum_{\mathcal{L}(G)} u^{n-m} K_m(G). \quad (2.4)$$

This is well defined by Proposition 2.2.

LEMMA 2.5. *As submodules of $k_*\{G\}$, we have that $R_* = Q_*$.*

Proof. Select an addition–contraction tree $\mathcal{T}(G)$ for G , and assume that each node H has successors $H \sqcup f$ and H/f . Summing over the nodes, we obtain

$$\sum_{\mathcal{T}(G)} u^{n-|\pi(H)|} (H - H \sqcup f - uH/f) = G - \sum_{\mathcal{L}(G)} u^{n-m} K_m(G),$$

which shows that each generator of Q_* lies in R_* .

On the other hand, $\mathcal{L}(G) = \mathcal{L}(G \sqcup e) \sqcup \mathcal{L}(G/e)$ for any e in $N(G)$, so that

$$\begin{aligned} G - G \sqcup e - uG/e &= \left(G - \sum_{\mathcal{L}(G)} u^{n-m} K_m(G) \right) - \left(G \sqcup e - \sum_{\mathcal{L}(G \sqcup e)} u^{n-m} K_m(G \sqcup e) \right) \\ &\quad - u \left(G/e - \sum_{\mathcal{L}(G/e)} u^{n-1-m} K_m(G/e) \right). \end{aligned}$$

This shows that each generator of R_* lies in Q_* , and hence yields the result.

COROLLARY 2.6. *As k_* modules, we have that T_* is isomorphic to the free module*

$$k_*\{K\} = k_*\{\emptyset, K_1, K_2, \dots\}$$

generated by all complete graphs.

Proof. Define an epimorphism $p: k_*\{G\} \rightarrow k_*\{K\}$ by means of $p(G) = \sum_{\mathcal{L}(G)} u^{n-m} K_m(G)$. Clearly p has kernel Q_* .

Following the lead of Yetter [17], we may interpret Corollary 2.6 neatly in terms of *rewrite rules* for elements of $k_*\{G\}$. Thus (2.3) specifies a scheme by which such elements may be successively rewritten, and the rule is readily seen to satisfy both *well foundedness* and *local confluence* (see [6, 7]). Thus each element has a *terminal form*, which is provided by the relations defining Q_* .

So far, we only considered $k_*\{G\}$ as a k_* -module. In fact it is richly endowed with further algebraic structure, which we now investigate.

We begin with the coproduct δ . This is the map $k_*\{G\} \rightarrow k_*\{G\} \otimes k_*\{G\}$ defined by

$$\delta(G) = \sum_{w \in V} G|_w \otimes G|_{w'}$$

on each generator G . It is routine to check that δ is coassociative, cocommutative and has counit ε given by

$$\varepsilon(G) = \begin{cases} 1 & \text{if } G = \emptyset, \\ 0 & \text{otherwise.} \end{cases}$$

By way of illustration, we record that the effect of the coproduct on the complete and null graphs is given by

$$\delta(K_n) = \sum_j \binom{n}{j} K_{n-j} \otimes K_j \quad \text{and} \quad \delta(N_n) = \sum_j \binom{n}{j} N_{n-j} \otimes N_j \quad (2.7)$$

for all $n \geq 1$.

Secondly, $k_*\{G\}$ admits a *delta operator*; in other words an endomorphism Δ^k which reduces grading by 2 and which interacts with the coproduct according to the rules

$$\delta \cdot \Delta^k = (\Delta^k \otimes 1) \cdot \delta = \delta \cdot (\Delta^k \otimes 1). \quad (2.8)$$

It is defined by

$$\Delta^k(G) = \sum_{w \in V}^{\#} u^{|w|-1} G|_{w'},$$

where $\sum^\#$ denotes summation over *non-empty, independent* subsets of V . The characterising property (2.8) is easily verified, since all three expressions have the common value

$$\sum_{X \subseteq W}^\# \sum_{W \subseteq V} u^{|X|-1} G|_{W \setminus X} \otimes G|_{V \setminus W}.$$

On the complete and null graphs, we have that

$$\Delta^k(K_n) = nK_{n-1} \quad \text{and} \quad \Delta^k(N_n) = \sum_{j=0}^{n-1} \binom{n}{j} u^{n-j-1} N_j \quad (2.9)$$

for all $n \geq 1$.

Thirdly, there is a product map $\mu: k_*\{G\} \otimes k_*\{G\} \rightarrow k_*\{G\}$ defined on generators by the rule $G \cdot H = \mu(G \otimes H) = G \sqcup H$. This is readily seen to be an associative and commutative coalgebra map, with identity $\emptyset$.

PROPOSITION 2.10. *The product interacts with Δ^k according to the formula*

$$\Delta^k(G \cdot H) = \Delta^k(G) \cdot H + G \cdot \Delta^k(H) + u \Delta^k(G) \cdot \Delta^k(H).$$

Proof. The right-hand expression arises by respectively splitting the non-empty independent subsets of $V(G \sqcup H)$ into those contained in $V(G)$, or in $V(H)$, or partly in each.

Fourthly, $k_*\{G\}$ admits an *antipode* S ; that is, an endomorphism which is a convolution inverse to the identity map. It is defined by

$$S(G) = \sum_{\pi(V)} (-1)^{|\pi|} |\pi|! G|_\pi.$$

The characterising property is a consequence of the identity

$$\sum_{W \subseteq V} \sum_{\pi(W)} (-1)^{|\pi|} |\pi|! G|_\pi \sqcup G|_{W'} = 0,$$

which is simply verified (for non-empty G) by enumerating the occurrences of each graph on the left-hand side.

By way of example, $S(N_n) = (-1)^n N_n$ for the null graph N_n . This follows by noting that $\sum_{\pi(n)} (-1)^{|\pi|} |\pi|! = (-1)^n$, a fact whose proof is a standard exercise with Stirling numbers.

We summarise the algebraic constructions of this section by saying that $k_*\{G\}$ is a *Hopf algebra with delta operator*. The aim of the next section is to show that the whole superstructure is preserved by the projection map p of Corollary 2.6.

3. The simple Tutte algebra

We now proceed to investigate how T_* inherits from $k_*\{G\}$ the structure of Hopf algebra with delta operator, and to identify the quotient map with a version of the classical chromatic polynomial.

The first of these aims is encapsulated in the following result, which refers back to (2.3).

THEOREM 3.1. *The submodule R_* is a Hopf ideal of $k_*\{G\}$ and is invariant under the action of Δ^k .*

Proof. There are four stages. First we explain why R_* is a coideal.

Choose a generator $G - G \sqcup e - uG/e$, where $e = vw$. In order to consider its image under the coproduct we divide the subsets of V into three distinct types, namely those which contain both v and w , those which contain only one, and those which contain neither. We give these the generic labels W_1 , W_2 and W_3 respectively, noting that the complement of W_1 is W_3 (and vice versa), whilst the complement of W_2 is also W_2 . We further observe that there is a natural one-to-one correspondence between the totality of subsets W_1 and W_3 , and the subsets of $V(G/e)$; it is convenient to use the same notation for a given subset and its image under this correspondence.

Thus we have ten equations

$$\begin{aligned} G \sqcup e|_{w_1} &= G|_{w_1} \sqcup e, & G \sqcup e|_{v \wedge w_1} &= G|_{v \wedge w_1}, \\ G \sqcup e|_{w_2} &= G|_{w_2}, & G \sqcup e|_{v \wedge w_2} &= G|_{v \wedge w_2}, \\ G \sqcup e|_{w_3} &= G|_{w_3}, & G \sqcup e|_{v \wedge w_3} &= G|_{v \wedge w_3} \sqcup e, \\ G/e|_{w_1} &= G|_{w_1}/e, & G/e|_{v \wedge w_1} &= G|_{v \wedge w_1}, \\ G/e|_{w_3} &= G|_{w_3}, & G/e|_{v \wedge w_3} &= G|_{v \wedge w_3}/e, \end{aligned}$$

which conspire to show that

$$\begin{aligned} \delta(G - G \sqcup e - uG/e) &= \sum_{w_1} (G|_{w_1} - G|_{w_1} \sqcup e - uG|_{w_1}/e) \otimes G|_{w_1} \\ &\quad + \sum_{w_3} G|_{w_3} \otimes (G|_{w_3} - G|_{w_3} \sqcup e - uG|_{w_3}/e). \end{aligned}$$

So $\delta(R_*)$ lies in $R_* \otimes k_*\{G\} + k_*\{G\} \otimes R_*$, whence R_* is a coideal.

The second stage is to prove that R_* is invariant under the action of Δ^k . Using the notation and equations introduced above, we obtain

$$\begin{aligned} \Delta^k(G - G \sqcup e - uG/e) &= \Delta^k(G) - \sum_{w_2}^{\#} u^{|w_2|-1} G|_{w_2} \\ &\quad - \sum_{w_3}^{\#} u^{|w_3|-1} G|_{w_3} - u \sum_{w_1}^{\#} u^{|w_1|-2} G|_{w_1} \\ &= 0. \end{aligned}$$

Thus R_* actually lies in the kernel of Δ^k .

The third stage is to prove that R_* is an ideal in $k_*\{G\}$. This is easily accomplished with the aid of the identity

$$(G - G \sqcup e - uG/e) \sqcup H = (G \sqcup H) - (G \sqcup H) \sqcup e - u(G \sqcup H)/e$$

where, as usual, G and H are generators of $k_*\{G\}$.

Finally, we must show that R_* is invariant under the action of the antipode S . Again we use the notation and equations above to establish the crucial identity

$$S(G - G \sqcup e - uG/e) = \sum (G|_{\pi} - G|_{\pi} \sqcup e - uG|_{\pi}/e),$$

where $e = vw$ and the summation ranges over all partitions π of $V(G)$ for which v and w lie in the *same* block. The contributions of the remaining partitions are restricted to G and $G \sqcup e$, and therefore cancel.

The proof is now complete.

We are thus in a position to elaborate on Corollary 2.6 by describing the extra structure maps induced on T_* ; we use the same notation for these maps as for their counterparts in $k_*\{G\}$.

THEOREM 3.2. *As a Hopf algebra, T_* is isomorphic to the binomial Hopf algebra $k_*[x]$, whose coproduct and antipode are specified respectively by*

$$\delta(x) = x \otimes 1 + 1 \otimes x \quad \text{and} \quad S(x) = -x.$$

Furthermore, the induced delta operator acts on any polynomial as the discrete derivative

$$\Delta^k f(x) = (f(x+u) - f(x))/u.$$

Proof. The relations defining Q_* show that, for all $n \geq 0$, the null and complete graphs are related in T_* by $N_n \equiv K_n$ modulo multiplicative decomposables. Thus we may write

$$T_* = k_*\{\emptyset, N_1, N_2, \dots\}.$$

Since the product structure stems from disjoint union, we deduce that $T_* = k_*[N_1]$ as algebras; it is therefore convenient to rewrite N_1 as x (and hence N_n as x^n).

Moreover (2.7) ensures that

$$\delta(x^n) = \sum \binom{n}{j} x^{n-j} \otimes x^j,$$

whilst our example of the action of the antipode reduces to $S(x^n) = (-1)^n x^n$. So T_* is indeed isomorphic to the binomial Hopf algebra.

To identify Δ^k , we remark that (2.9) may now be recast as

$$\Delta^k(x^n) = \sum_{j=0}^{n-1} \binom{n}{j} u^{n-j-1} x^j = ((x+u)^n - x^n)/u. \quad (3.3)$$

Thus Δ^k acts as the discrete derivative on the standard basis for the polynomial algebra, and hence on all polynomials.

It is a simple fact that the discrete derivative interacts with a product of polynomials according to the rule

$$\begin{aligned} (f(x+u)g(x+u) - f(x)g(x))/u &= ((f(x+u) - f(x))/u)g(x) \\ &\quad + f(x)(g(x+u) - g(x))/u \\ &\quad + (f(x+u) - f(x))(g(x+u) - g(x))/u, \end{aligned} \quad (3.4)$$

as prophesied by Proposition 2.10.

Our second aim in this section is to identify the projection $q: k_*\{G\} \rightarrow k_*[x]$ provided by Theorem 3.2. Before so doing, we observe that (2.9) provides us with the formula $\Delta^k(q(K_n)) = nq(K_{n-1})$, for all $n \geq 1$. This in turn tells us that the sequence of polynomials $q(K_n)$ is the *associated sequence* for the delta operator Δ^k . So by referring to [13] (or by simple computation), we obtain

$$q(K_n) = x(x-u) \dots (x-(n-1)u) \quad \text{for } n \geq 0. \quad (3.5)$$

To analyse q , we invoke a homogenised version of the classical chromatic polynomial $\chi(G; x)$ (for example, see [2]) of a graph G . It is defined for any G on n vertices by replacing each term x^k in $\chi(G; x)$ by $u^{n-k}x^k$. Thus the *homogeneous chromatic polynomial* $\chi^k(G; x)$ lies in the ring $\mathbb{Z}[u][x] \cong \mathbb{Z}[u, x]$. It may be characterised by the fact that for positive integers t , the substitution $x = ut$ yields the number of colourings of G on up to t colours, multiplied by the factor u^n .

By way of example, we note that the expressions

$$\chi^k(N_n; x) = x^n \quad \text{and} \quad \chi^k(K_n; x) = x(x - u) \dots (x - (n - 1)u) \quad (3.6)$$

follow from the definitions.

Completing our analysis is now a formality.

THEOREM 3.7. *The projection q maps a generator G of $k_*\{G\}$ to the polynomial $\chi^k(G; x)$.*

Proof. By virtue of (2.4), we have that

$$q(G) = \sum_{\mathcal{L}(G)} u^{n-m} q(K_m).$$

On the other hand, Proposition 2.2 shows that

$$\chi^k(G; x) = \sum_{\mathcal{L}(G)} u^{n-m} \chi^k(K_m; x),$$

so the result follows by combining (3.5) with (3.6).

We should also remark on the following alternative characterisation of the homogeneous chromatic polynomial, which in spite of being more complicated will be important in later work.

Let $S_{n,j}$ be the Stirling number of the second kind [1], which by definition enumerates the partitions of the set $\{1, \dots, n\}$ into j parts, for $1 \leq j \leq n$. When the substitution

$$x^n = \sum_j u^{n-j} S_{n,j}[t]_j = u^{-t} \sum_s \binom{t}{s} (su)^n (u - 1)^{t-s} \quad (3.8)$$

is applied to $\chi^k(G; x)$, it yields $\sum_j u^{n-j} c(t, j)$, where $c(t, j)$ denotes the number of colourings of G using exactly j of t colours. To verify that this agrees with our earlier characterisation, it is in fact sufficient to check the complete graphs. This we leave to the reader; details will appear in [3].

It would, of course, be possible to present the material of the last two sections in the reverse order, by *defining* the projection q in terms of the homogeneous chromatic polynomial and then investigating the resulting properties. It is purely a matter of taste as to which approach is preferred.

4. Graphs and partitions

We now generalise the constructions of §2 so as to take account of the partition of the vertices of a graph induced by contracting non-edges. To this end, we introduce the polynomial algebra $\Phi_* = \mathbb{Z}[\phi_1, \phi_2, \dots]$, where each generator

ϕ_i is assumed to be of grading $2i$, and ϕ_0 is taken to be 1. Then given a partition $\pi = \{\pi(1), \dots, \pi(k)\}$ of a finite set, we define its *type* $\tau(\pi)$ in Φ_* to be the monomial

$$\tau(\pi) = \prod_i \phi_{|\pi(i)|-1}.$$

Note that there is an epimorphism $\gamma: \Phi_* \rightarrow k_*$ which is defined for all $i \geq 0$ by $\gamma(\phi_i) = u^i$. Following [9], we label γ as the *genus*.

Now take the free module $\Phi_*\{G\}$ generated by all graphs G (including $\emptyset$), and let G be of grading $2|V|$. We propose to impose certain relations on $\Phi_*\{G\}$ involving addition and contraction of non-edges. These will be consistent under the genus with those already imposed on $k_*\{G\}$ in § 2.

Thus, by analogy with (2.4) we let Q_*^Φ denote the submodule generated by all terms of the form

$$G - \sum_{\mathcal{L}(G)} \tau_m K_m(G), \quad (4.1)$$

where each τ_m denotes the type of the partition $\pi(K_m(G))$. The terms (4.1) are well defined by reference to Proposition 2.2.

We cannot define a straightforward analogue of (2.3) based on a *single* addition and contraction operation, and we have therefore lost the attractive interpretation in terms of simple rewrite rules. This situation can be remedied by working more generally with *numbered* graphs, each of whose vertices is assigned a positive integer; however, we shall save this option for the future, since it introduces further complexities elsewhere.

We write T_*^Φ for the quotient module $\Phi_*\{G\}/Q_*^\Phi$, which is graded since the terms in (4.1) are all homogeneous. We refer to T_*^Φ as the *umbral Tutte algebra*; its module structure is readily determined.

PROPOSITION 4.2. *As Φ_* modules, we have that T_*^Φ is isomorphic to the free module*

$$\Phi_*\{K\} = \Phi_*\{\emptyset, K_1, K_2, \dots\}$$

generated by all complete graphs.

Proof. Define an epimorphism $p: \Phi_*\{G\} \rightarrow \Phi_*\{K\}$ by means of $p(G) = \sum_{\mathcal{L}(G)} \tau_m K_m$. Clearly p has kernel Q_*^Φ .

Note that the genus passes to the obvious map $T_*^\Phi \rightarrow T_*$ of quotients.

Following § 2 we now outline how $\Phi_*\{G\}$ may be considered as a Hopf algebra with delta operator in such a way that the map $\Phi_*\{G\} \rightarrow k_*\{G\}$, induced by the genus, preserves all the structure.

The coproduct δ is the map $\Phi_*\{G\} \rightarrow \Phi_*\{G\} \otimes \Phi_*\{G\}$ defined by

$$\delta(G) = \sum_{w \in V} G|_w \otimes G|_w.$$

on each generator G , and the counit ε is given by

$$\varepsilon(G) = \begin{cases} 1 & \text{if } G = \emptyset, \\ 0 & \text{otherwise.} \end{cases}$$

The delta operator Δ^ϕ is defined by

$$\Delta^\phi(G) = \sum_{w \subseteq V}^\# \phi|_{w|-1} G|_w, \quad (4.3)$$

where we recall that $\sum^\#$ denotes summation over non-empty, independent subsets of V . The characterising property (2.8) is easily verified, since the three relevant expressions have the common value

$$\sum_{x \subseteq W}^\# \sum_{w \subseteq V} \phi|_{x|-1} G|_{w \setminus x} \otimes G|_{v \setminus w}.$$

On the complete and null graphs, we have that

$$\Delta^\phi(K_n) = nK_{n-1} \quad \text{and} \quad \Delta^\phi(N_n) = \sum_{j=0}^{n-1} \binom{n}{j} \phi_{n-j-1} N_j \quad (4.4)$$

for all $n \geq 1$.

The product map $\mu: \Phi_*\{G\} \otimes \Phi_*\{G\} \rightarrow \Phi_*\{G\}$ is much more subtle. As before, we denote its action on generators by $G.H = \mu(G \otimes H)$, and its definition certainly involves the disjoint union $G \sqcup H$; however, extra input is required.

We first recall from [12] the poset $\mathcal{A}(G)$ of *admissible* partitions; the blocks of such partitions either contain an edge of G or are singletons. We refer to an admissible partition in $\mathcal{A}(G \sqcup H)$ as *mixed* if every non-singleton block contains vertices of both G and H , including the *trivial* mixed, or *unmixed* partition whose blocks are all singletons. For each mixed ρ we form the quotient $G \sqcup H / \rho$, by choosing vertices to be the set of blocks ρ and edges $\rho(i)\rho(j)$ to be those pairs of blocks whose elements support at least one edge of $G \sqcup H$ between them. Finally, we add an edge between each non-singleton block of ρ and every other vertex, omitting duplicate edges as necessary. We label the resulting graph $M_\rho(G, H)$, and refer to it as a *mixture* of G and H ; if ρ is trivial, we set $M_\rho(G, H)$ to be the *unmixture* $G \sqcup H$.

To each mixture we assign a *type* $\tau_\rho = \tau_\rho(G, H)$, which is a homogeneous polynomial in Φ_* of degree $|V(G)| + |V(H)| - |\rho|$, and which is computed in terms of admissible partitions as follows. Let $\Gamma(\rho)$ denote the set of all chains

$$\gamma = \{\pi_0 < \dots < \pi_i < \dots < \pi_t < \rho\}$$

which start at $\pi_0 = 0$ (the partition into singletons) and for which each partition π_i is *unmixed*. Then we define

$$\tau_\rho = \sum_{\Gamma(\rho)} (-1)^t \tau(\rho/\pi_t) \prod_i \tau(\pi_i/\pi_{i-1}), \quad (4.5)$$

where π_i/π_{i-1} denotes the partition of π_{i-1} induced by π_i . We let τ_ρ be 1 if ρ is unmixed. Note that for each mixed partition ρ , there is a unique maximal element ρ° in the poset of unmixed partitions $\{\pi: \pi < \rho\}$; it is obtained from ρ by suitably splitting every mixed block into unmixed parts.

The following result about such types will be helpful later.

LEMMA 4.6. *For any decomposition $\rho = \sigma \sqcup \sigma'$, we may compute $\tau_\rho(G, H)$ by restricting attention to chains of the form*

$$\gamma = \{0 < \pi_1 < \dots < \pi_s < \pi_s \sqcup \pi'_1 < \dots < \pi_s \sqcup \pi'_t < \rho\}, \quad (4.7)$$

where the non-trivial blocks of π_i and π'_j lie entirely in σ and σ' respectively.

Proof. Take an arbitrary chain γ in $\Gamma(\rho)$, and consider the first occurrence (if any) of a string of the form

$$\{\dots < \pi_i \sqcup \pi'_j < \pi_i \sqcup \pi'_{j+1} < \pi_{i+1} \sqcup \pi'_{j+1} < \dots\}.$$

Replacing this string by $\{\dots < \pi_i \sqcup \pi'_j < \pi_{i+1} \sqcup \pi'_{j+1} < \dots\}$ yields a second chain in $\Gamma(\rho)$ whose contribution to τ_ρ is easily seen to differ only in sign from that of γ (for their lengths differ by one). The contributions therefore cancel, and the only chains not eliminated by repeated application of this procedure are of the stated form.

We may now describe the product operation. We define

$$G \cdot H = G \sqcup H + \sum'_{\rho} \tau_{\rho} M_{\rho}(G, H) = \sum_{\rho} \tau_{\rho} M_{\rho}(G, H), \quad (4.8)$$

where $\sum'$ denotes summation over all non-trivial mixtures of G and H , and $\sum$ includes the unmixture. By way of the simplest example, note that $N_n \cdot N_m = N_{n+m}$ for all $n, m \geq 0$.

It is clear that this operation is commutative, and that the empty graph $\emptyset$ acts as the identity. However, associativity demands more thought, and hinges on the observation that both $F \cdot (G \cdot H)$ and $(F \cdot G) \cdot H$ may be rewritten as

$$F \cdot G \cdot H = \sum_{\sigma} \tau_{\sigma} M_{\sigma}(F, G, H), \quad (4.9)$$

where σ ranges over all elements of $\mathcal{A}(F \sqcup G \sqcup H)$ which (unless trivial), have all their non-singleton blocks containing vertices from two or more of F , G , and H . The mixtures $M_{\sigma}(F, G, H)$ are constructed accordingly, and the types τ_{σ} are defined in terms of chains whose elements (except for the last) are unmixed.

To verify (4.9), we first note that $F \cdot (G \cdot H)$ is a sum of terms

$$\tau_{\nu} \tau_{\rho} M_{\nu}(F, M_{\rho}(G, H)), \quad (4.10)$$

where ρ lies in $\mathcal{A}(G \sqcup H)$ and ν lies in $\mathcal{A}(F \sqcup M_{\rho}(G, H))$. In fact, to compute τ_{ν} we may apply Lemma 4.6 and only consider chains of the form (4.7), where π_i lies in $\mathcal{A}(M_{\rho}(G, H))$ and π'_j lies in $\mathcal{A}(F)$.

Suppose that ρ is non-trivial, consider a term of the form (4.10), and take chains γ_1 contributing to τ_{ρ} and γ_2 of the above type contributing to τ_{ν} . Pair these with the term $\tau_{\nu} M_{\nu}(F, G \sqcup H)$ and the chain contributing to τ_{ν} obtained by concatenating γ_1 and γ_2 (where γ_1 is interpreted as leaving the singletons of $V(F)$ unaltered). These two combinations make contributions to $F \cdot (G \cdot H)$ differing only in sign, and so may be cancelled; all remaining terms have the form

$$\{0 < \pi_1^G \sqcup \pi_1^H < \dots < \pi_s^G \sqcup \pi_s^H < \pi_1^F \sqcup \pi_s^G \sqcup \pi_s^H < \dots < \pi_t^F \sqcup \pi_s^G \sqcup \pi_s^H\},$$

where π_i^F , π_j^G and π_k^H lie in $\mathcal{A}(F)$, $\mathcal{A}(G)$ and $\mathcal{A}(H)$ respectively. Modulo a final application of Lemma 4.6 (with $\sigma = V(F)$), we have thus demonstrated that $F \cdot (G \cdot H)$ is given by the right-hand side of (4.9). The argument is concluded by applying similar reasoning to $(F \cdot G) \cdot H$.

We must also show that our product is a coalgebra map; for this we need some notation. Each mixture $M_{\rho}(G, H)$ has the blocks of ρ as its vertices, so for any decomposition $\rho = \sigma \sqcup \sigma'$ we have induced subgraphs $M_{\rho}(G, H)|_{\sigma}$ and $M_{\rho}(G, H)|_{\sigma'}$. Now the blocks of σ are made up from a subset S of $V(G)$ and a subset T of $V(H)$, whilst the blocks of σ' are made up from the complementary

subsets S' and T' , thereby giving rise to induced subgraphs $G|_S, H|_T, G|_{S'}$ and $H|_{T'}$ (up to two of which may be empty). Hence there are also mixtures $M_\sigma(G|_S, H|_T)$ and $M_{\sigma'}(G|_{S'}, H|_{T'})$; moreover, at most one of these can be the unmixture.

It follows from these definitions that

$$M_\rho(G, H)|_\sigma = M_\sigma(G|_S, H|_T) \quad \text{and} \quad M_\rho(G, H)|_{\sigma'} = M_{\sigma'}(G|_{S'}, H|_{T'}). \quad (4.11)$$

Moreover, the corresponding types satisfy

$$\tau_\rho(G, H) = \tau_\sigma(G|_S, H|_T) \tau_{\sigma'}(G|_{S'}, H|_{T'}),$$

by appealing to Lemma 4.6 again. Consequently, the coproduct on $G \cdot H$ may be rewritten

$$\begin{aligned} \sum_\rho \tau_\rho \sum_{\sigma \subseteq \rho} M_\rho(G, H)|_\sigma \otimes M_\rho(G, H)|_{\sigma'} &= \left(\sum_\sigma \tau_\sigma M_\sigma(G|_S, H|_T) \right) \\ &\quad \otimes \left(\sum_{\sigma'} \tau_{\sigma'} M_{\sigma'}(G|_{S'}, H|_{T'}) \right), \end{aligned}$$

which is the product $\delta(G) \cdot \delta(H)$. Thus the product commutes with δ .

The genus collapses the product in $\Phi_*\{G\}$ onto the product in $k_*\{G\}$, because each type τ_ρ reduces to zero under the substitution of u^i for ϕ_i , given any $i \geq 1$. To see this, we simply pair together chains in $\Gamma(\rho)$ of the form

$$\{\pi_0 < \pi_1 < \dots < \pi_t < \rho\} \quad \text{and} \quad \{\pi_0 < \pi_1 < \dots < \pi_t < \rho^\circ < \rho\},$$

whose contributions $(-1)^t u^d$ and $(-1)^{t+1} u^d$ (where $d = |V(G)| + |V(H)| - |\rho|$) cancel each other out.

To study the interaction of the product and the delta operator on $\Phi_*\{G\}$, we must make a definition. The homogeneous element

$$\alpha_{i,j} \in \Phi_{2i+2j-2}, \quad \text{for } i, j \geq 0, \quad (4.12)$$

is defined to be the type of the maximal partition $1_{i,j}$ in $\mathcal{A}(K_i \sqcup K_j)$. Of course $1_{i,j}$ is mixed, since it contains all the vertices in a single block. Careful scrutiny of (4.12) reveals that

$$\alpha_{1,0} = \alpha_{0,1} = 1 \quad \text{and} \quad \alpha_{i,0} = \alpha_{0,i} = 0 \quad \text{for all } i > 0;$$

also $\alpha_{i,j} = \alpha_{j,i}$ for all i, j .

We can now state

PROPOSITION 4.13. *The product on $\Phi_*\{G\}$ interacts with Δ^ϕ according to the formula*

$$\Delta^\phi(G \cdot H) = \sum_{i,j \geq 0} \alpha_{i,j} (\Delta^\phi)^i(G) \cdot (\Delta^\phi)^j(H) / i! j!. \quad (4.14)$$

Proof. The left-hand expression may be expanded as

$$\Delta^\phi \left(\sum_\rho \tau_\rho M_\rho(G, H) \right) = \sum_\rho \tau_\rho \left\{ \sum_{\sigma \subseteq \rho}^\# \phi_{|\sigma|-1} M_\rho(G, H)|_\sigma \right\},$$

recalling that σ' is the complement of σ in ρ . The fact that σ is independent means that it is either a single block $\rho(t)$ of ρ , or else consists entirely of

singletons of $V(G \sqcup H)$. In other words, from (4.11),

$$M_\rho(G, H)|_{\sigma'} = M_{\sigma'}(G|_{S'}, H|_{T'}),$$

where $\tau_\rho = \phi_{|\rho(t)|-1}\tau_{\sigma'}$ or $\tau_{\sigma'}$ in the respective cases. We deduce that

$$\tau_\rho \phi_{|\sigma|-1} M_\rho(G, H)|_{\sigma'} = \tau_{\sigma'} \phi_{|S \sqcup T|-1} M_{\sigma'}(G|_{S'}, H|_{T'}) \quad (4.15)$$

in both cases.

Now consider the right-hand side of (4.14). Each term $(\Delta^\Phi)^i(G)/i!$ may be written

$$\sum_{w_1, \dots, w_i}^\# \phi_{|w_1|-1} \dots \phi_{|w_i|-1} G|_W / i!,$$

where W is the disjoint union of the independent subsets $W_1, \dots, W_i$. Since this disjoint union occurs in all possible orders, we may remove the factor $i!$ on condition that we consider each decomposition of W as *unordered* (that is, as a partition ω , say, of type $\phi(\omega)$), and count it only once. With these conventions, the right-hand side is made up of terms of the form

$$\alpha_{i,j} \tau_\rho \phi(\omega) \phi(\nu) M_\rho(G|_W, H|_X), \quad (4.16)$$

where $\omega = \{W_1, \dots, W_i\}$ and $\nu = \{X_1, \dots, X_j\}$ are partitions of W and X respectively.

Observe that (4.15) accounts for exactly those terms of (4.16) for which ω and ν consist entirely of singletons (and therefore have type 1), and for which ϕ_{i+j-1} is the element selected from $\alpha_{i,j}$. We thus have to prove that the other terms cancel amongst themselves. This is accomplished by pairing terms of

$$\alpha_{i,j} (\Delta^\Phi)^i(G) \cdot (\Delta^\Phi)^j(H) / i! j! \quad \text{and} \quad \alpha_{|W|, |X|} (\Delta^\Phi)^{|W|}(G) \cdot (\Delta^\Phi)^{|X|}(H) / i! j!$$

as follows. Take the contribution to the former by some chain γ of $\Gamma(1_{i,j})$, and partitions ω and ν ; it differs only in sign from the contribution to the latter by the chain of $\Gamma(1_{|W|, |X|})$ obtained by concatenating $\{0 < \omega \sqcup \nu\}$ and γ (where γ is interpreted blockwise), and the partitions into singletons.

The remaining details are now straightforward.

Note that the formula of (4.14) reduces to that of Proposition 2.10 under the genus, since we have already established that all non-trivial types reduce to zero.

Finally, $\Phi_*\{G\}$ admits the antipode S defined by

$$S(G) = \sum_{\pi(V)} (-1)^{|\pi|} |\pi|! G|_{\pi(1)} \dots G|_{\pi(k)},$$

where $k = |\pi|$. The proof of the characterising property parallels that in §2 for the antipode of $k_*\{G\}$, to which S reduces under the genus. Similarly, the formula $S(N_n) = (-1)^n N_n$ still holds for the null graph N_n .

5. The umbral Tutte algebra

We now proceed to investigate how T_*^Φ inherits from $\Phi_*\{G\}$ the structure of Hopf algebra with delta operator, and to identify the quotient map with the umbral chromatic polynomial.

Referring back to (4.1), our aim is to emulate Theorem 3.1 by proving the following.

THEOREM 5.1. *The submodule Q_*^Φ is a Hopf ideal of $\Phi_*\{g\}$, and is invariant under the action of Δ^Φ .*

Proof. As before, there are four stages, and we first explain why Q_*^Φ is a coideal.

Choose a generator $G = \sum_{\mathcal{L}(G)} \tau_m K_m$. To each decomposition $V(G) = W \sqcup W'$ and pair of colour partitions σ of $G|_W$ and σ' of $G|_{W'}$, there corresponds a colour partition π of G and a decomposition $\pi = \sigma \sqcup \sigma'$. Clearly $\tau(\pi) = \tau(\sigma)\tau(\sigma')$. Therefore the coproduct acts such that

$$\delta\left(\sum_{\mathcal{L}(G)} \tau_m K_m\right) = \sum_{W \subseteq V(G)} \left(\sum_{\mathcal{L}(G|_W)} \tau_i K_i\right) \otimes \left(\sum_{\mathcal{L}(G|_{W'})} \tau_j K_j\right).$$

Thus we may write the image of the chosen generator under δ as

$$\sum_{W \subseteq V(G)} \left\{ \left(G|_W - \sum_{\mathcal{L}(G|_W)} \tau_i K_i\right) \otimes G|_{W'} + \left(\sum_{\mathcal{L}(G|_W)} \tau_i K_i\right) \otimes \left(G|_{W'} - \sum_{\mathcal{L}(G|_{W'})} \tau_j K_j\right) \right\}.$$

So $\delta(Q_*^\Phi)$ lies in $Q_*^\Phi \otimes \Phi_*\{G\} + \Phi_*\{G\} \otimes Q_*^\Phi$, whence Q_*^Φ is a coideal.

The second stage is to prove that Q_*^Φ is invariant under the action of Δ^Φ . Referring again to the generator $G = \sum_{\mathcal{L}(G)} \tau_m K_m$, we recall from (4.3) and (4.4) that we have

$$\Delta^\Phi(G) = \sum_{W \subseteq V}^\# \phi_{|W|-1} G|_W, \quad \text{and} \quad \Delta^\Phi(K_m) = m K_{m-1}.$$

Since W is independent, each colour partition σ' of $G|_{W'}$ into $m-1$ parts may be extended to a colour partition π of G into m parts, and $\tau(\pi) = \phi_{|W|-1} \tau(\sigma')$. Moreover, the same π will arise in m different ways, according to which block is omitted in σ' . Therefore we have that Δ^Φ acts on the generator to yield

$$\sum_{W \subseteq V}^\# \phi_{|W|-1} \left(G|_{W'} - \sum_{\mathcal{L}(G|_{W'})} \tau_j K_j\right),$$

which visibly still lies in Q_*^Φ .

The third stage is to prove that Q_*^Φ is an ideal in $\Phi_*\{G\}$, which involves showing that

$$G \cdot H - \sum_{\mathcal{L}(G)} \tau_m K_m(G) \cdot H$$

lies in Q_*^Φ for all G and H . It is thus sufficient to verify that

$$\sum_{\mathcal{L}(G \cdot H)} \tau_i K_i = \left(\sum_{\mathcal{L}(G)} \tau_m\right) \left(\sum_{\mathcal{L}(K_m(G) \cdot H)} \tau_j K_j\right). \quad (5.2)$$

The left-hand side is a sum of terms of the form $\tau_\rho \tau_k K_k$, where ρ ranges over the mixed partitions of $\mathcal{A}(G \sqcup H)$, whilst the right-hand side is a sum of terms of the form $\tau_m \tau_\sigma \tau_l K_l$, where σ ranges over the mixed partitions of $\mathcal{A}(K_m(G) \sqcup H)$. Thus the left-hand terms agree with those right-hand terms which arise when both $\tau_m = 1$ (and therefore $K_m = K_{|G|}$) and also the chains contributing to τ_σ contain no independent blocks of $V(G)$. To verify (5.2), we must therefore show that the remaining right-hand terms cancel amongst themselves.

Such cancellation is effected as follows. Let π be a colour partition of G , of type τ_m , and let γ be a chain contributing to τ_σ . By Lemma 4.6, we may assume

that γ has the form

$$\{\pi_1^G < \dots < \pi_s^G < \pi_s^G \sqcup \pi_1^H < \dots < \pi_s^G \sqcup \pi_t^H < \sigma\}.$$

First assume that π is trivial, and that π_r^G (with $r \geq 2$) contains the first occurrence of an independent block of $V(G)$. Write $\pi_r^G = \pi_r(D) \sqcup \pi_r(I)$ for its decomposition into respectively dependent, and independent, sets of blocks, noting that $\pi_{r-1}(I)$ consists entirely of singletons. If $\pi_{r-1}(D) \neq \pi_r(D)$, then we pair γ with the chain obtained by substituting

$$\pi_{r-1}^G < \pi_{r-1}(I) \sqcup \pi_r(D) < \pi_r(I) \sqcup \pi_r(D)$$

for $\pi_{r-1}^G < \pi_r^G$. These two chains make contributions to the right-hand side of (5.2) which cancel, leaving unaccounted for only those with either π trivial, $r = 2$ and $\pi_2(D)$ empty, or π non-trivial. But these also cancel by pairing each of the former type with one of the latter, chosen such that π is $\pi_2(I)$.

Finally, we must show that $Q_\star^\Phi$ is invariant under the action of the antipode S . A direct combinatorial proof of this fact is surprisingly intricate, and it is most simply verified by an unilluminating induction based on the defining property of S . We leave the details to the reader, together with the comment that the most revealing approach is introduced below, at the end of the section.

The proof is complete.

We may now mimic §3, and elaborate on Proposition 4.2 by describing the extra structure maps induced on $T_\star^\Phi$; we use the same notation for these maps as for their counterparts in $\Phi_\star\{G\}$.

THEOREM 5.3. *As a Hopf algebra, $T_\star^\Phi$ is isomorphic to the binomial Hopf algebra $\Phi_\star[x]$, whose coproduct and antipode are specified respectively by*

$$\delta(x) = x \otimes 1 + 1 \otimes x \quad \text{and} \quad S(x) = -x.$$

Furthermore, the induced delta operator acts on any polynomial as the umbral difference operator (see [9])

$$\Delta^\Phi f(x) = f(x + \phi) - f(x), \quad \phi^i \equiv \phi_{i-1},$$

and all structure maps are compatible under the genus with those of $k_\star[x]$.

Proof. The relations defining $Q_\star^\Phi$ show that, for all $n \geq 0$, the null and complete graphs are related in $T_\star^\Phi$ by $N_n \equiv K_n$ modulo multiplicative decomposables. Thus we may write

$$T_\star^\Phi = \Phi_\star\{\emptyset, N_1, N_2, \dots\}.$$

Since the product structure restricted to null graphs stems from disjoint union, we deduce that $T_\star^\Phi = \Phi_\star[N_1]$ as algebras; it is therefore convenient to rewrite N_1 as x (and hence N_n as x^n).

In these terms we have that $\delta(x^n) = \sum \binom{n}{j} x^{n-j} \otimes x^j$ whilst the action of the antipode reduces to $S(x^n) = (-1)^n x^n$. So $T_\star^\Phi$ is indeed isomorphic to the binomial Hopf algebra.

To identify Δ^Φ , we remark that (4.4) may now be recast as

$$\Delta^\Phi(x^n) = \sum_{j=0}^{n-1} \binom{n}{j} \phi_{n-j-1} x^j = (\phi + x)^n - x^n, \quad \phi^i \equiv \phi_{i-1}. \quad (5.4)$$

Thus Δ^ϕ acts as the umbral difference operator on the standard basis for the polynomial algebra, and hence on all polynomials.

Finally, our comments on the action of the genus at the pertinent points of § 4 ensure that it is compatible with all the structures under discussion.

As explained in [8], the umbral difference operator acts on a product of polynomials according to a rule

$$\Delta^\phi(f(x)g(x)) = \sum_{i,j \geq 0} a_{i,j} (\Delta^\phi)^i f(x) (\Delta^\phi)^j g(x)$$

for certain $a_{i,j}$ in $\Phi_* \otimes \mathbb{Q}$. With the help of (4.14), we can now deduce that $i! j! a_{i,j} = \alpha_{i,j}$.

Our second aim in this section is to identify the projection $q^\phi: \Phi_*\{G\} \rightarrow \Phi_*[x]$ provided by Theorem 5.3. Before so doing, we note that the formula $\Delta^\phi(q^\phi(K_n)) = n \Delta^\phi(q^\phi(K_{n-1}))$, for all $n \geq 1$, follows from (4.4).

This in turn tells us that the sequence of polynomials $q^\phi(K_n)$ is the associated sequence for the delta operator Δ^ϕ . So by referring to [8] (or by direct computation), we deduce that $q^\phi(K_n)$ is the n th conjugate Bell polynomial $b_n^\phi(x)$, for which no simple closed formula (other than as provided by (5.8) below) seems to appear in the literature.

To analyse q^ϕ , we invoke the umbral chromatic polynomial of [12], denoted by $\chi^\phi(G; x)$ for any graph G . It is a homogeneous element of the ring $\Phi_*[x]$, has degree $|V(G)|$, and is characterised by the fact that for positive integers t the umbral substitution $x = t\phi$ yields the expression $\sum \tau(\pi)$, summed over the colour partitions defined by all colourings of G on up to t colours. It is not difficult to see that the formulae

$$\chi^\phi(N_n; x) = x^n \quad \text{and} \quad \chi^\phi(K_n; x) = b_n^\phi(x) \quad (5.5)$$

follow from the definitions; these are explained in [12]. It is also straightforward to verify that application of the genus to $\chi^\phi(G; x)$ yields the homogeneous chromatic polynomial $\chi^k(G; x)$ of § 3, and to the umbral substitution $x = t\phi$ yields the umbral substitution of (3.8).

We can now complete our analysis.

THEOREM 5.6. *The projection q^ϕ maps a generator G of $\Phi_*\{G\}$ to the polynomial $\chi^\phi(G; x)$.*

Proof. By virtue of (4.1), we have that

$$q^\phi(G) = \sum_{\mathcal{L}(G)} \tau_m q^\phi(K_m).$$

On the other hand, Proposition 2.2 shows that

$$\chi^\phi(G; x) = \sum_{\mathcal{L}(G)} \tau_m \chi^\phi(K_m; x),$$

so the result follows from (5.5).

We conclude this section by outlining a deeper approach to Theorem 5.6, which involves further work in understanding the module $\mathcal{Q}_*^\phi$ but which has the advantage that it yields an explicit formula for $\chi^\phi(G; x)$ in terms of the poset $\mathcal{A}(G)$ of admissible partitions.

To each σ in $\mathcal{A}(G)$ we associate the null graph $N_{|\sigma|} = N_{|\sigma|}(G)$ and the *Möbius* type μ_σ (see [9]), which is computed as follows. Let $C(\sigma)$ denote the set of chains

$$\gamma = \{\sigma_0 < \dots < \sigma_i < \dots < \sigma_t\}$$

in $\mathcal{A}(G)$, which start at $\sigma_0 = 0$ and end at $\sigma_t = \sigma$. Then we define

$$\mu_\sigma = \sum_{C(\sigma)} (-1)^t \prod_i \tau(\sigma_i / \sigma_{i-1}).$$

Thus μ_σ is a homogeneous polynomial in Φ_* of degree $|V(G)| - |\sigma|$.

Then by following the lead of several of the above proofs and analysing contributing chains, we can establish the following directly from the definitions.

PROPOSITION 5.7. *The terms of the form*

$$G - \sum_{\mathcal{A}(G)} \mu_\sigma N_{|\sigma|}(G)$$

provide an alternative set of generators for the module Q_^Φ . Furthermore, we have that*

$$\sum_{\mathcal{A}(G \cdot H)} \mu_\sigma N_{|\sigma|}(G \cdot H) = \left(\sum_{\mathcal{A}(G)} \mu_\pi N_{|\pi|}(G) \right) \left(\sum_{\mathcal{A}(H)} \mu_\rho N_{|\rho|}(H) \right),$$

for arbitrary graphs G and H .

This result may be used to give a simplified proof that Q_*^Φ is an ideal which is invariant under the antipode, and provides a clearer understanding of the product operation in $\Phi_*\{G\}$. In tandem with Theorem 5.6, it yields a demonstration of the formula

$$\chi^\Phi(G; x) = \sum_{\mathcal{A}(G)} \mu_\sigma x^{|\sigma|} \quad (5.8)$$

of [12].

6. Formal group theory

To conclude, we outline how our preceding work on algebras of graphs may be linked with the theory of formal groups. Not only do these links allow us to construct further examples of such algebras, and to simplify their theory, but they also suggest the possibility of working in reverse. This is of interest because several aspects of formal group theory, such as the proof of Lazard's theorem [4], have a distinctly combinatorial flavour, and it has long been a challenge to develop methods with the potential to provide explicit explanation in terms of graphs, posets or other discrete structures. We hope to return to this aspect of our investigations elsewhere.

Recall that a one-dimensional *formal group law* over a commutative ring R with identity is a formal power series $f = f(X, Y)$ in $R[[X, Y]]$ satisfying the identities

$$\begin{aligned} f(X, Y) &\equiv X + Y \pmod{(X, Y)^2}, \\ f(X, Y) &= f(Y, X), \\ f(X, f(Y, Z)) &= f(f(X, Y), Z). \end{aligned}$$

We shall write $f(X, Y) = \sum_{i,j} b_{i,j} X^i Y^j$, noting that as a result of the above identities,

$$b_{1,0} = b_{0,1} = 1 \quad \text{and} \quad b_{i,0} = b_{0,i} = 0 \quad \text{for all } i > 0;$$

also $b_{i,j} = b_{j,i}$ for all i, j . We shall always assume that R is free of additive torsion, so that f defines and is defined by its *exponential* series $\exp(Z)$. This is a formal power series $\sum_i e_i Z^i$ in $R \otimes \mathbb{Q}[[Z]]$, which admits a compositional inverse $\log(Z)$ (thus $e_0 = 0$, and we may take $e_1 = 1$) and which is uniquely determined by the rule

$$\exp(\log(X) + \log(Y)) = f(X, Y). \quad (6.1)$$

It is also convenient (and makes no difference to the theory) to insist that $R = R^*$ is *graded* and that X, Y and f have grading 2. Thus $b_{i,j}$ and e_i have gradings $2 - 2i - 2j$ and $2 - 2i$ respectively. We recall that the graded dual of R^* over itself is conventionally written as $R = R_*$, denoting that the grading is negated.

The following is a straightforward consequence of these definitions.

LEMMA 6.2. *Each element $\psi_i = i! e_i$ lies in R_* . Moreover, if we write Ψ_* for the subring of R_* generated by $\psi_1, \psi_2, \dots$, then each element $\beta_{i,j} = i! j! b_{i,j}$ lies in Ψ_* .*

We may thus rewrite the formal group law and its exponential series as

$$f(X, Y) = \sum_{i,j} \beta_{i,j} X^i Y^j / i! j! \quad \text{and} \quad \exp(Z) = \sum_i \psi_i Z^i / i!.$$

In this setting, we can now make explicit the relationship between the formal group law and the corresponding delta operator Δ^ψ over $\Psi_*[x]$.

Consider the Hopf algebra $\Psi^*((D))$ of formal divided power series, where D has grading 2 and the coproduct and antipode are given respectively by $\delta(D) = D \otimes 1 + 1 \otimes D$ and $D \mapsto -D$. Then the exponential series, and hence the formal group law, is an element $W = \exp(D)$ whose divided powers given an alternative set of generators; that is,

$$\Psi^*((W)) = \Psi^*((D)). \quad (6.3)$$

Thus by (6.1), since $D = \log(W)$, the formal group law merely describes the action of the coproduct on W as

$$\delta(W) = \sum_{i,j} \beta_{i,j} \frac{W^i}{i!} \otimes \frac{W^j}{j!}.$$

We now dualise as graded Hopf algebras over Ψ^* , so that $\Psi^*((D))$ becomes the binomial Hopf algebra $\Psi_*[x]$. The duality is conveniently expressed by interpreting D as d/dx , so that

$$\langle D^i / i! \mid x^j \rangle = (d/dx)^i x^j / i! \big|_{x=0} = \delta_{i,j},$$

whence we see that $\Psi^*((D))$ acts on its dual as the algebra of formal differential operators applied to polynomial functions. Hence the exp series, and therefore the formal group law, may be expressed as

$$\Delta^\psi = D + \psi_1 \frac{D^2}{2!} + \dots + \psi_{i-1} \frac{D^i}{i!} + \dots;$$

in other words as a delta operator.

The above action is a special case of the following construction, applicable to any Hopf algebra A_* , and its dual A^* over the ring of scalars R_* . Consider $f \in A^*$ as a functional $f: A_* \rightarrow R_*$; then it acts on A_* by the composition

$$A_* \xrightarrow{\delta} A_* \otimes A_* \xrightarrow{f \otimes 1} R_* \otimes A_* \longrightarrow A_*, \quad (6.4)$$

where the final map is multiplication by the scalars.

In the case under consideration, where A_* is $\Psi_*[x]$ and R_* is Ψ_* , we deduce that the action of Δ^ψ (which as a functional satisfies $x^i \mapsto \psi_{i-1}$) on polynomials $p(x)$ may be succinctly expressed as the umbral difference operator

$$p(x) \mapsto p(\psi + x) - p(x), \quad \psi^i \equiv \psi_{i-1}.$$

That Δ^ψ interacts with δ according to (2.8) follows immediately from (6.4).

This whole procedure is reversible in that, given any delta operator over a ring free of additive torsion, we may at once recover the associated formal group law by means of (6.1).

Two other aspects of this correspondence are important. Firstly, we remark that (6.3) provides us, by duality, with an alternative basis of polynomials $b_n^\psi(x)$ for $\Psi_*[x]$, which from the point of view of Δ^ψ make up the associated sequence, and are transparently binomial. Secondly, to describe the action of Δ^ψ on a product of polynomials, we apply (6.4) and compute as follows:

$$\begin{aligned} \Delta^\psi p(x)q(x) &= \Delta^\psi \mu(p(x) \otimes q(x)) = \mu\{\delta(\Delta^\psi)(p(x) \otimes q(x))\} \\ &= \mu\left\{\left(\sum_{i,j} \beta_{i,j}(\Delta^\psi)^i \otimes (\Delta^\psi)^j / i! j!\right)(p(x) \otimes q(x))\right\} \\ &= \sum_{i,j} \beta_{i,j}(\Delta^\psi)^i p(x) (\Delta^\psi)^j q(x) / i! j!, \end{aligned} \quad (6.5)$$

where μ is the product map. Thus the formal group law reappears as the *Leibniz rule* for applying the differential operator to a product.

Let us briefly mention some standard examples.

First is the graded *multiplicative* formal group law, given over $k_* = \mathbb{Z}[u]$ by

$$f(X, Y) = X + Y + uXY \quad \text{and} \quad \exp(Z) = (e^{uZ} - 1)/u. \quad (6.6)$$

Here $\Psi_* = k_*$, and the corresponding delta operator is the discrete derivative Δ^k of (3.3).

Second is the *universal* formal group law, given over an infinitely generated polynomial algebra U_* (the Lazard ring) by

$$f(X, Y) = \sum_{i,j} a_{i,j} X^i Y^j \quad \text{and} \quad \exp(Z) = \sum_i b_i Z^i \quad (6.7)$$

for certain elements $a_{i,j}$ and b_i in U_* . Here $\Psi_* = \Phi_*$ is a *proper* subalgebra of U_* (generated by the terms $\phi_i = i! b_i$), and the corresponding delta operator is the umbral difference operator Δ^ϕ of (5.4). This example is universal in the sense that any other formal group law is its homomorphic image.

Third is the *additive* formal group law, which is the simplest of all. It is given over $\mathbb{Z}$ by

$$f(X, Y) = X + Y \quad \text{and} \quad \exp(Z) = Z, \quad (6.8)$$

and $\Psi_* = \mathbb{Z}$. The corresponding delta operator is just the derivative d/dx .

We can now state our main result, using the above notation.

THEOREM 6.9. *Given any formal group law f , we may construct a projection of Hopf algebras with delta operator*

$$q^\psi: \Psi_*\{G\} \rightarrow \Psi_*[x],$$

where the delta operators have the Leibniz rule

$$\Delta^\psi(G \cdot H) = \sum_{i,j} \beta_{i,j} (\Delta^\psi)^i(G) \cdot (\Delta^\psi)^j(H) / i! j!$$

for generators G and H of $\Psi_*\{G\}$. In addition, $q^\psi(G)$ may be interpreted as a chromatic polynomial of G , enumerating colourings in terms of types defined by the elements ψ_i .

Proof. Algebraically, we make the required construction simply by forming the tensor product over Φ_* of the projection q^ϕ of Theorem 5.3 with the identity on Ψ_* . Here Ψ_* is given the structure of a Φ_* -module by means of the homomorphism defined by $\phi_i \mapsto \psi_i$, for all $i \geq 1$. However, it is worth detailing the combinatorial interpretation of this procedure.

The underlying principle is that the type of any partition $\pi = \{\pi(1), \dots, \pi(k)\}$ of a finite set is now defined to be the monomial

$$\tau(\pi) = \prod_i \psi_{|\pi(i)|-1} \quad (6.10)$$

in Ψ_* . We then begin with the free module $\Psi_*\{G\}$, and follow §4 by introducing the submodule of relations Q_*^ψ , generated by the elements $G - \sum_{\mathcal{A}(G)} \tau_m K_m(G)$. The quotient is the free module $\Psi_*\{K\}$ generated by all complete graphs. Continuing to follow §4, we impose the structure of Hopf algebra with delta operator (which we shall label temporarily as $\Delta_{\mathcal{C}}^\psi$) on $\Psi_*\{G\}$ by employing the formula corresponding to (4.3). As in §5, we see that Q_*^ψ is then a Hopf ideal, and may identify the quotient more precisely as the binomial Hopf algebra $\Psi_*[x]$. Similarly, $\Delta_{\mathcal{C}}^\psi$ reduces to the umbral difference operator Δ^ψ , and its associated sequence is represented by the images $q^\psi(K_n)$ of the complete graphs under the projection.

To interpret $q^\psi(G)$ as a chromatic polynomial $\chi^\psi(G; x)$, we duplicate the arguments concluding §5. The upshot is that, for positive integers t , the umbral substitution of $x = t\psi$ in $\chi^\psi(G; x)$ yields the expression $\sum \tau(\pi)$ (recalling (6.10)), summed over the colour partitions induced by all colourings of G on up to t colours.

Clearly Δ^ψ has the stated Leibniz rule on $\Psi_*[x]$, by application of (6.5); it remains only to explain why the corresponding formula holds for $\Delta_{\mathcal{C}}^\psi$. For this it is convenient first to introduce the dual of the Hopf algebra $\Psi_*\{G\}$, which we label as $\Psi^*\{\{g\}\}$ and which is the free Ψ^* -module whose basis elements consist of all characteristic functionals g , such that $g(G) = \delta_{G,H}$ for any graphs G and H . Note that certain infinite sums are allowed, as the notation is meant to imply. The structure maps of this Hopf algebra are given in a straightforward fashion by dualisation. Thus, for example, its product and coproduct maps are described respectively by

$$g \cdot h = \sum_j j \quad \text{and} \quad \delta(j) = \sum \tau_\rho g \otimes h,$$

where the first summation is over all graphs J and subsets $W \subseteq V(J)$ such that $J|_W = G$ and $J|_{W^c} = H$, whilst the second is over all graphs G, H and mixtures $\rho \in \mathcal{A}(G \sqcup H)$ such that $J = M_\rho(G, H)$.

We note that the iterated product formula $n_1^i = i! n_i$ follows directly from the definitions of the characteristic functionals of the null graphs.

Now consider the Hopf algebra monomorphism

$$r^\psi: \Psi^*((D)) \rightarrow \Psi^*\{\{g\}\}$$

dual to q^ψ . The composition

$$\Psi_*\{G\} \xrightarrow{q^\psi} \Psi_*[x] \longrightarrow \Psi_*,$$

where the second map is the functional associated to Δ^ψ , sends each generator G to $\chi^\psi(G; \psi)$. Since the only graph on i vertices which admits a colouring with a single colour is N_i , we deduce that $r^\psi(\Delta^\psi)$ is the functional specified by

$$G \mapsto \begin{cases} \psi_{i-1} & \text{if } G = N_i, \\ 0 & \text{otherwise.} \end{cases}$$

Therefore by (6.4), the action of $r^\psi(\Delta^\psi)$ is given by

$$r^\psi(\Delta^\psi)(G) = \sum_{W \subseteq V}^{\#} \psi_{|W|-1} G|_W$$

on the generators of $\Psi_*\{G\}$.

We conclude from (4.3) that $r^\psi(\Delta^\psi)$ may be identified with Δ_G^ψ ; but Δ^ψ and its image under r^ψ necessarily have the same Leibniz rule, whence our conclusion follows.

Note that it is now legitimate to return to our former convention, and abbreviate Δ_G^ψ to Δ^ψ .

The subtlety and interest of the situation described in Theorem 6.9 lie in choosing the elements ψ_i (which, as illustrated by (6.6), need not be algebraically independent) in such a way that $\tau(\pi)$ encodes information of significant combinatorial content.

We commend to the reader the study of the alternative set of generators for Q^ψ provided by the terms $G - \sum_{\mathcal{A}(G)} \mu_\sigma N_{|\sigma|}(G)$ (see the conclusion to § 5), and the instructive problem of explicitly computing the primitive element $r^\psi(D)$ in $\Psi^*\{\{g\}\}$.

We may finally return to the standard examples introduced above.

Firstly, the graded multiplicative formal group law (6.6) corresponds to our introductory work in § 2 and § 3, so that the resulting chromatic polynomial corresponds to our homogeneous version of the classical example. We note in passing that the *ungraded* multiplicative formal group law gives rise to the classical chromatic polynomial itself.

Secondly, the universal formal group law (6.7) corresponds to our main development in § 4 and § 5. The resulting umbral chromatic polynomial maps into all other such examples, and in particular the map into the multiplicative case is the genus.

The additive formal group law is an interesting but degenerate case which we have not so far considered, and for which we write $\psi = z$; thus $z^i \equiv 1$ for $i = 0$,

and 0 otherwise. The Hopf algebra $\mathbb{Z}\{G\}$ has the product map $G \cdot H = G \sqcup H$, and the delta operator D acts on G by

$$D(G) = \sum_{v \in V} G|_{V \setminus \{v\}},$$

so that $D(G \cdot H) = D(G) \cdot H + G \cdot D(H)$. The ideal Q^z is in this case generated by the rewrite rule $G = G \sqcup e$ for every non-edge e , so that the quotient Tutte algebra is $\mathbb{Z}[x]$, and the chromatic polynomial $\chi^z(G; x)$ is merely $x^{|V(G)|}$. It is instructive to verify that $\chi^z(G; tz)$ reduces to 0 if $t < |V(G)|$, and to

$$\binom{t}{|V(G)|}$$

otherwise, so that the polynomial enumerates colourings whose monochromatic blocks are singletons.

There are many other important formal group laws whose graph-theoretic significance is as yet unclear. For example, the p -typical cases look particularly promising, and are currently under investigation in [3]. We conclude with the more romantic and speculative suggestion that the laws related to elliptic curves may also have a rôle to play. The most classical such example lies over the ring $\mathbb{Z}[\frac{1}{2}][d, e]$, and is furnished by the following formula due to Euler:

$$f(X, Y) = \frac{x\sqrt{R(y)} + y\sqrt{R(x)}}{1 - ex^2y^2},$$

where $R(X) = 1 - 2dX^2 + eX^4$. In this case the logarithm is given by the formal elliptic integral

$$\log(Z) = \int_0^Z \frac{1}{\sqrt{R(t)}} dt,$$

which may be written as an inverse Jacobi elliptic function.

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